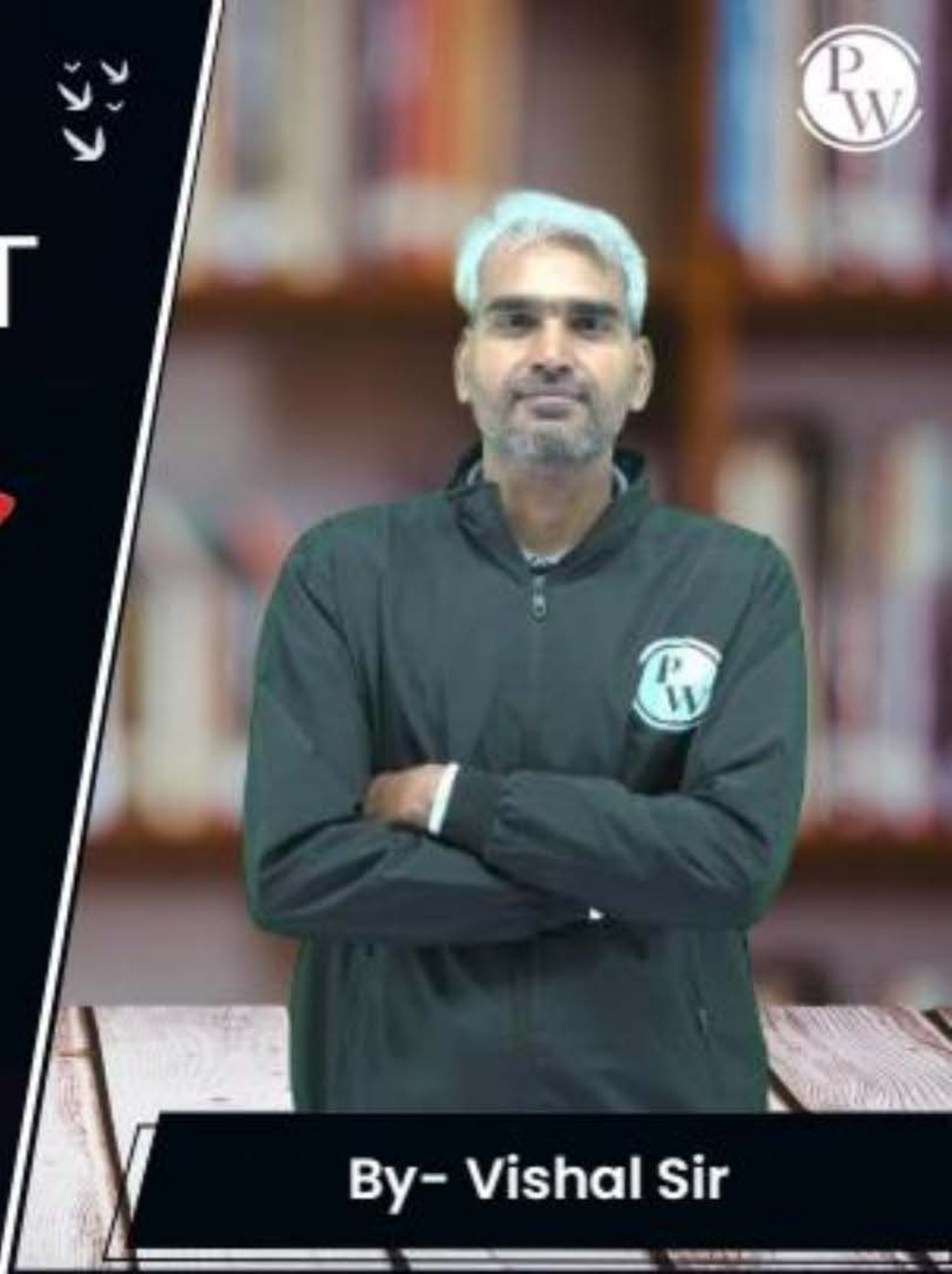
Computer Science & IT



Compiler Design

Syntax Analysis (Parser)

Lecture No. 11



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Construction of DFA of LR(0) items
- **Topic** LR(0) Parser and LR(0) grammar
- **Topic** LR Parsing algorithm

Slide

Topics to be Covered



Topic

SLR(1) Parser

Slide



Topic : DFA Of LR(0) items

Construct the DFA of LR(0) items for the following grammar.

1. $S \rightarrow AA$
   $A \rightarrow aA$
   $A \rightarrow b$

Already done

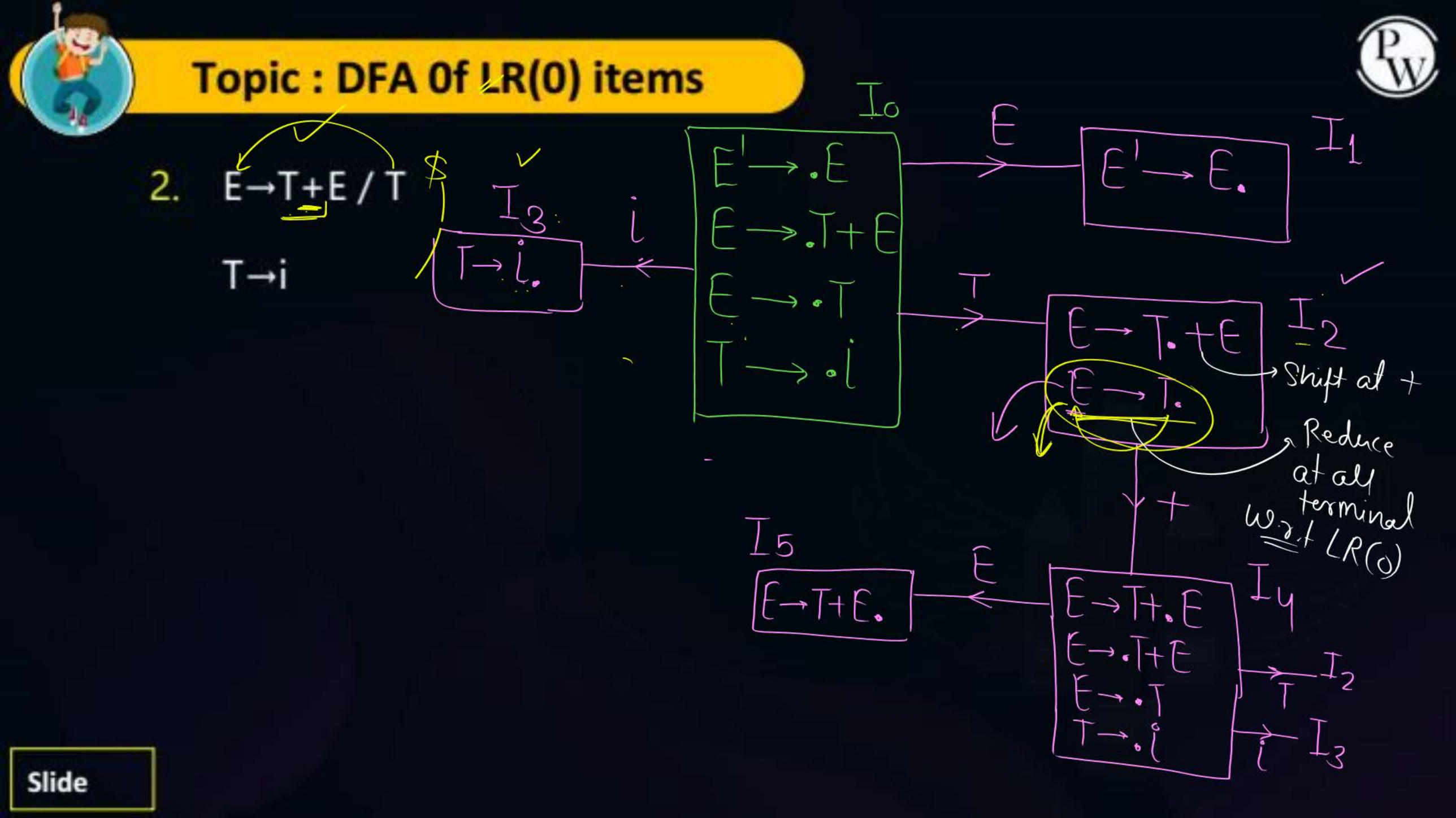
LR(0) grammar

Slide

Topic : DFA Of LR(0) items

2. $E \rightarrow T+E \mid T$

$T \rightarrow i$



Slide



Topic : SLR(1) Parser

* In LR(0) parser no look ahead is used to place the reduce entry in the action part of LR(0) parsing table, therefore there may be some unnecessary reduce entry in action part of LR(0) Parsing table, to avoid this problem SLR(1) can be used

* SLR(1) parser will use the lookahead of length '1' {i.e. Follow} to place the reduce entry in SLR(1) Parsing table

Slide



Topic : SLR(1) Parser

* SLR(1) Parser will use the same DFA of LR(0) items, but SRR(1) parser table may be different from LR(0) Parser table w.r.t reduce entries

SLR(1) Parser:

1. Obtain the augmented grammar
2. Construct the DFA of LR(0) items
3. Construct the SLR(1) parser table using the specified Rules {w.r.t Action & Goto Part}
4. If there is no Conflict in any entry of SLR(1) Parsing table, then grammar is SLR(1) grammar

Slide



Topic : Action part of SLR(1) Parser Table

1. Shift Action : If state "i" of the dfa includes an item of the form "$A \rightarrow \alpha . a \beta$", such that a is a terminal symbol and transition out of state "i" on input symbol a is to state "j", then

   ACTION [i , a] = Shift to state j = S_j

   (Same as LR(0))

2. Reduce Action : If state "i" of the dfa includes an item of the form "$A \rightarrow \alpha .$" where $A \rightarrow \alpha$ is the jth production of grammar, then

   ACTION [i , n] = Rj , $\forall n \in FOLLOW(A)$

   (Different from LR(0))

3. Accept : If state "i" of the dfa includes an item of the form "$S' \rightarrow S .$" , then

   ACTION [i , $] = Accept

   (Same as LR(0))

4. Error : Set all other entries of Action part as Error.

Slide



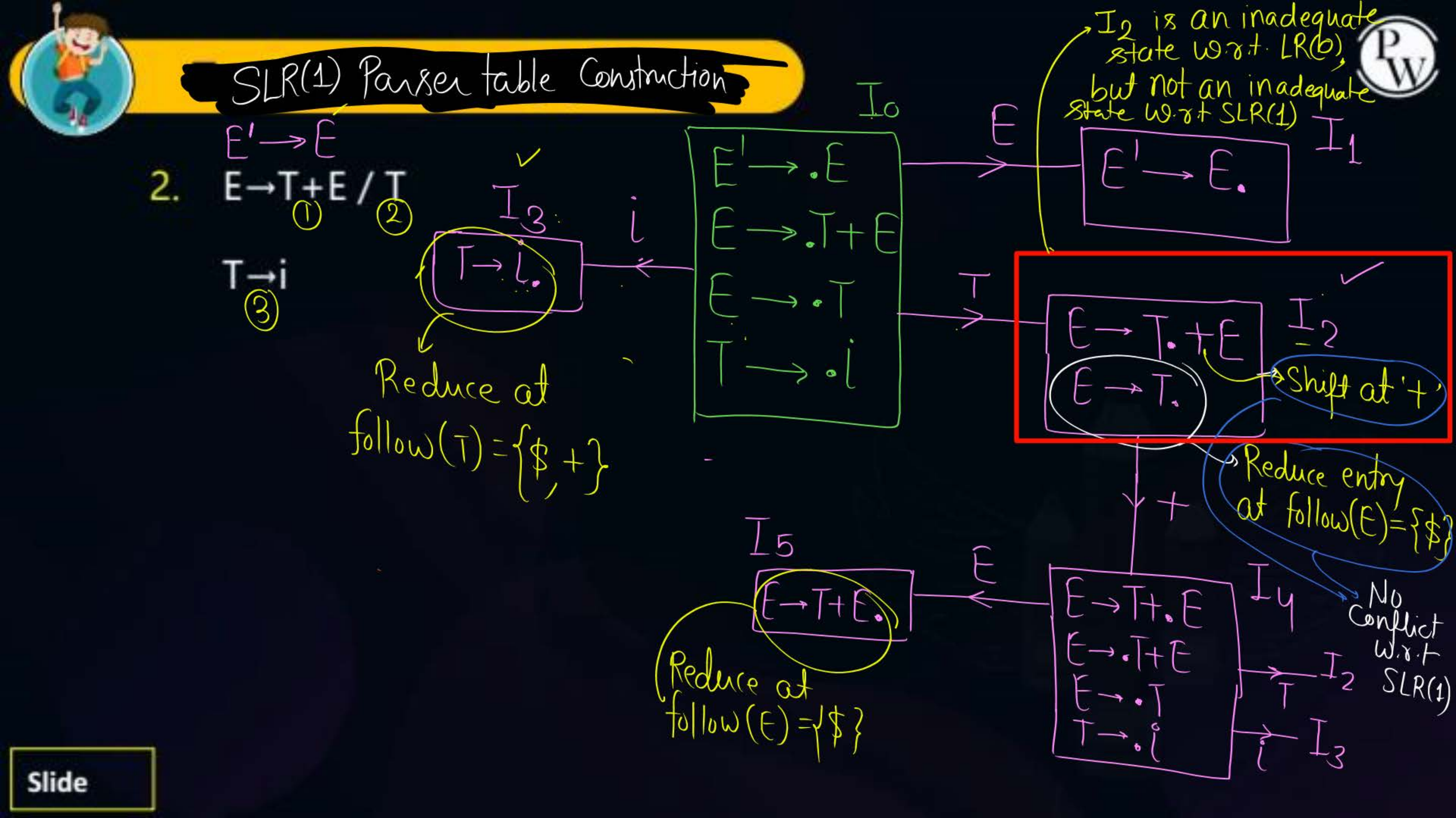
Topic : Goto part of SLR(1) Parser Table

- GOTO part is common for both LR(0) Parser and SLR(1) Parser.
- If there exists a transition out of state "i" on non-termina X to state "j" , then

  GOTO [i , X] = j

Slide

SLR(1) Parser table Construction



Slide



Topic : SLR(1) Parser table consturction

$$\begin{cases} E \rightarrow T+E \,/\, T \\ T \rightarrow i \end{cases}$$

	Action			Goto	
	i	+	$	E	T
0	S_3			1	2
1			Accept		
2		S_4	$R_2^{(E \rightarrow T)}$		
3		R_3	R_3		
4	S_3			5	2
5			R_1		

Slide



Topic : SLR(1) grammar

* If there is conflict in any entry of SLR(1) Parser table, then corresponding grammar is SLR(1) grammar,
  * If any entry in Action part of SLR(1) Parser table contains more than One option {SR or RR} then grammar is not SLR(1) grammar.

Slide

Topic : Conflict w.r.t. SLR(1) Parser

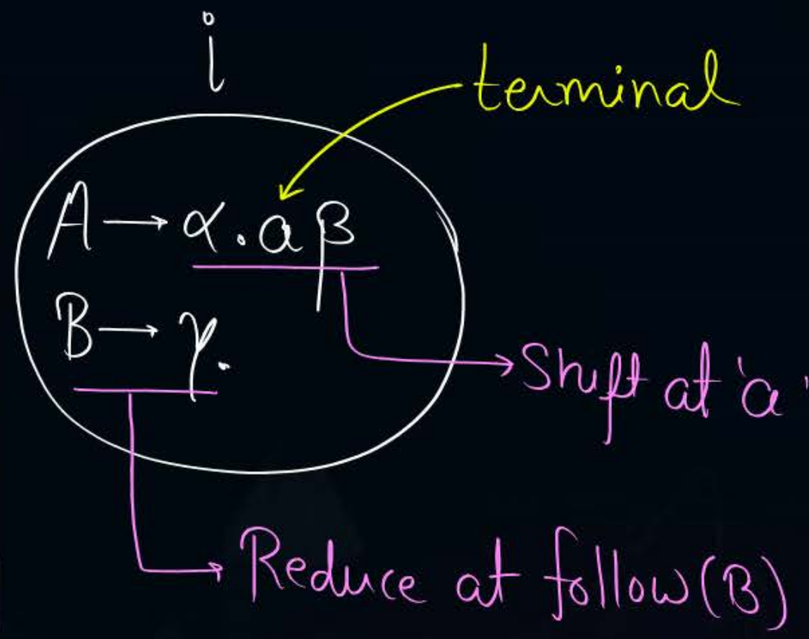
#Shift-Reduce Conflict w.r.t. SLR(1) Parser

Shift-Reduce Conflict exists in SLR(1) Parser table only if there exists a state 'i' in the DFA that contains items of the form

$A \rightarrow \alpha . a \beta$

& $B \rightarrow \gamma .$

and $a \in follow(B)$



SR Conflict exists in state 'i' only if

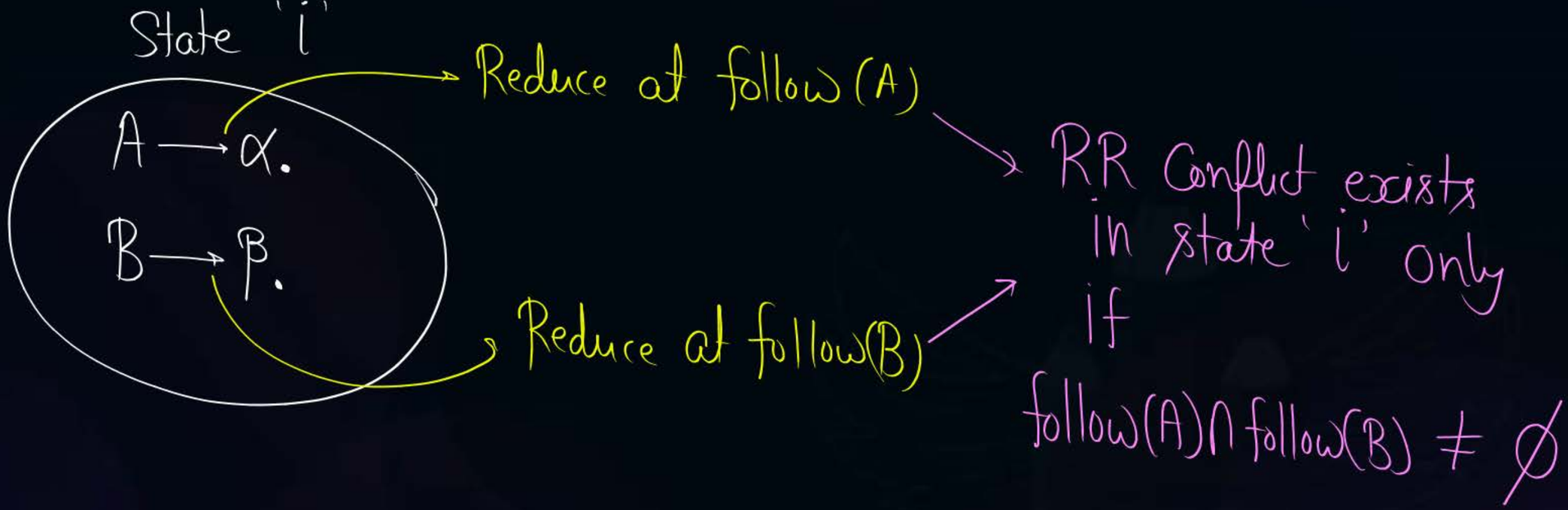
$a \in follow(B)$

Slide



Topic : Conflict w.r.t. SLR(1) Parser

#Reduce-Reduce Conflict w.r.t. SLR(1) Parser



Slide



Topic : Inadequate state w.r.t. SLR(1) Parser

Any state of the DFA with SR or RR Conflict w.r.t. SLR(1) parser is called inadequate state w.r.t. SLR(1) parser

Slide



Topic : LR Parsing Algorithm w.r.t. SLR(1) Parser

LR Parsing algo will remain same for all LR parser

Slide

* Every LR(0) grammar is SLR(1), but

Every SLR(1) grammar need not be LR(0)



Topic : DFA Of LR(0) items

Check whether the gramma is LR(0) or not? If not LR(0) then Check for SLR(1).

3. $E \rightarrow E+T \,/\, T$

$T \rightarrow i$

Left Recursion

∴ Not LL(1)

$E' \rightarrow E$
$E \rightarrow E+T$
$E \rightarrow T$
$T \rightarrow i$

I_0: $E' \rightarrow .E$, $E \rightarrow .E+T$, $E \rightarrow .T$, $T \rightarrow .i$

I_1: $E' \rightarrow E.$, $E \rightarrow E.+T$ — Accept at $

I_2: $E \rightarrow T.$

I_3: $T \rightarrow i.$

I_4: $E \rightarrow E+.T$, $T \rightarrow .i$

I_5: $E \rightarrow E+T.$

Shift at '+'

{No Reduce}

{No Reduce}

Accept & Shift Can never Create Conflict

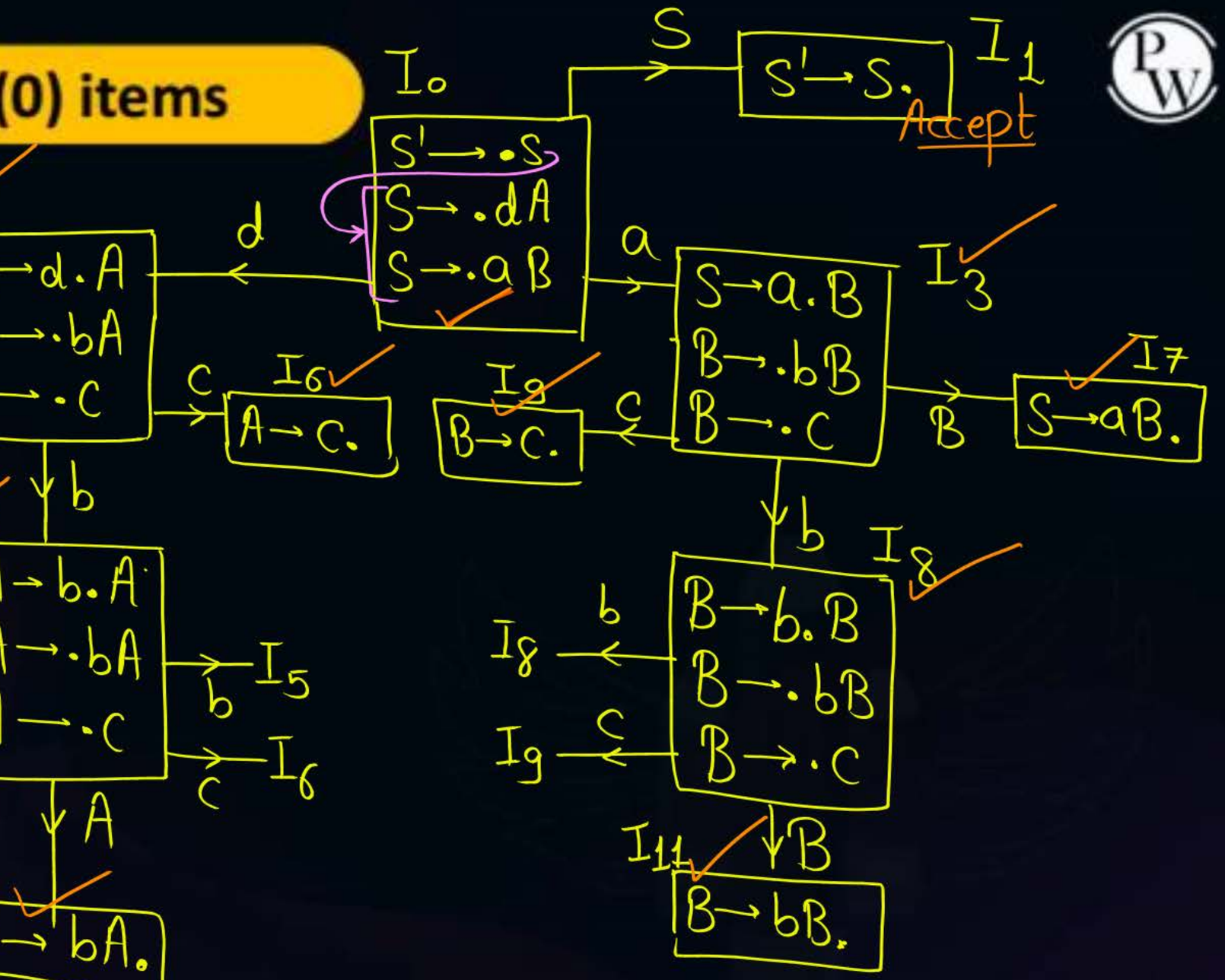
{ No SR or RR Conflict w.r.t. LR(0) ∴ LR(0) gramma as well as SLR(1) }

Slide

Topic : DFA Of LR(0) items

4. S→dA / aB
   A→bA / c
   B→bB / c

S'→S
S→dA
S→aB
A→bA
A→c
B→bB
B→c



No SR or RR Conflict w.r.t. LR(0) ∴ LR(0) gramma

Slide



Topic : DFA Of LR(0) items

H.W.

5. S→(L) /a

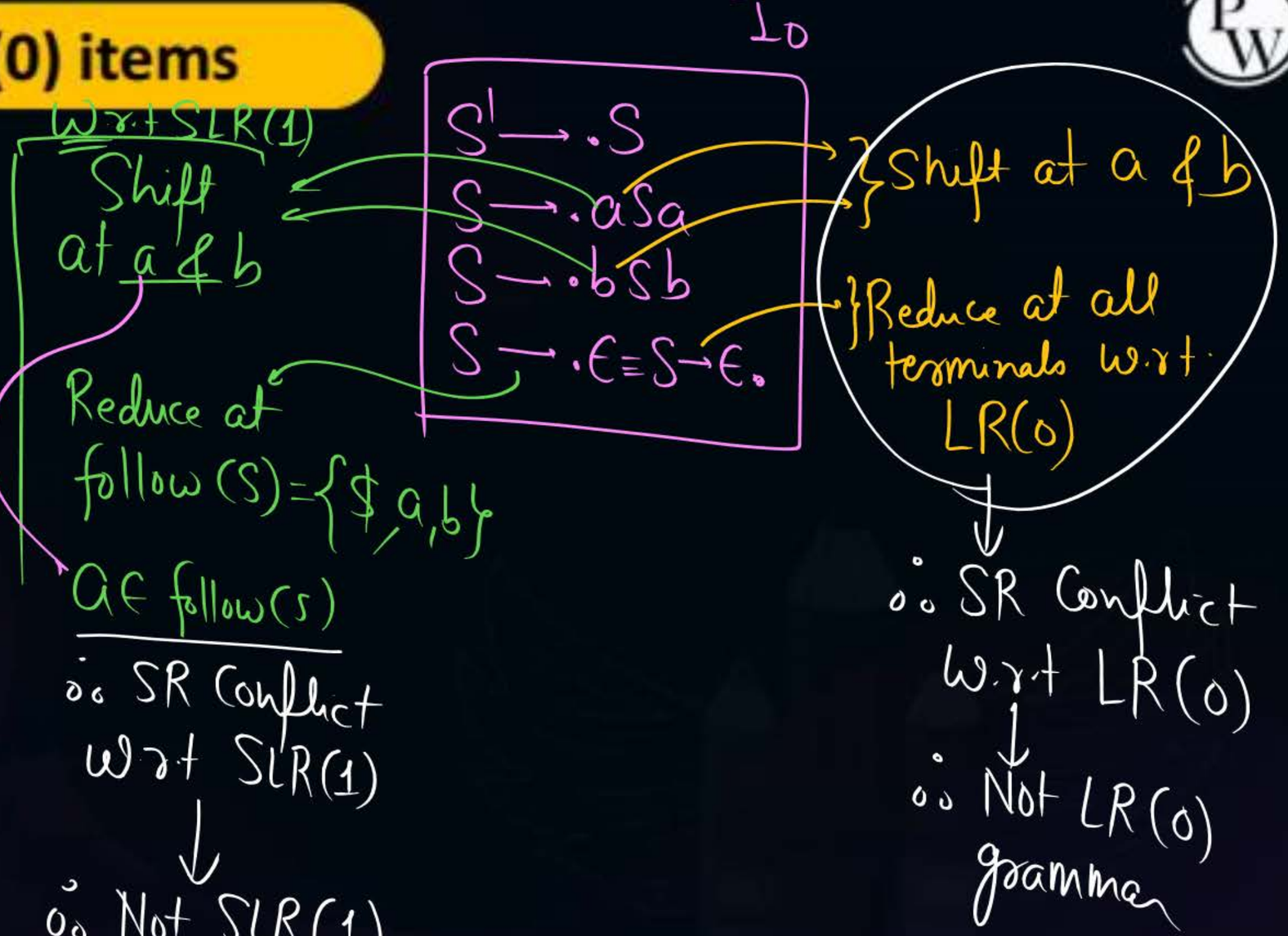
L→L , S / S

Slide

Topic : DFA Of LR(0) items

6. $S \rightarrow aSa / bSb / \epsilon$

$S' \rightarrow S$
$S \rightarrow aSa$
$S \rightarrow bSb$
$S \rightarrow \epsilon$



Slide



Topic : DFA Of LR(0) items

7. S→aSbS / bSaS / ϵ

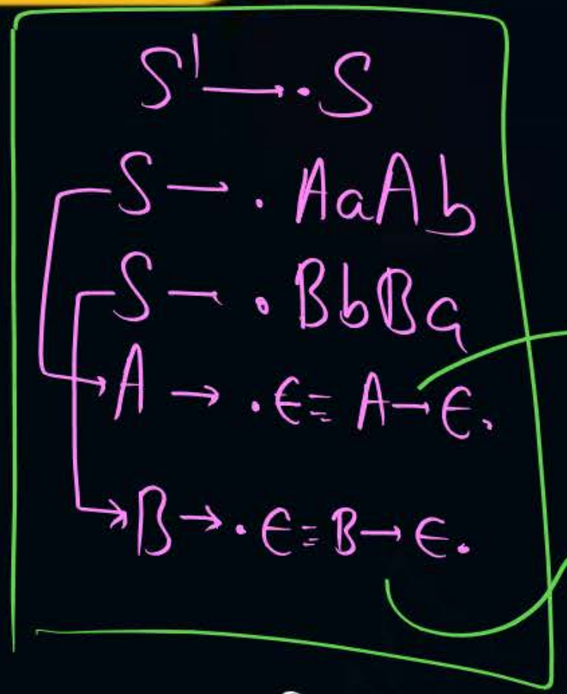
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Topic : DFA Of LR(0) items

8. S→AaAb /BbBa

   A→ ϵ

   B→ ϵ



Reduce at follow(A) = {a, b}

Reduce at follow(B) = {a, b}

follow(A) ∩ follow(B) = {a, b} ≠ ∅

∴ RR Conflict w.r.t SLR(1) {and Hence w.r.t LR(0) as well}

Slide

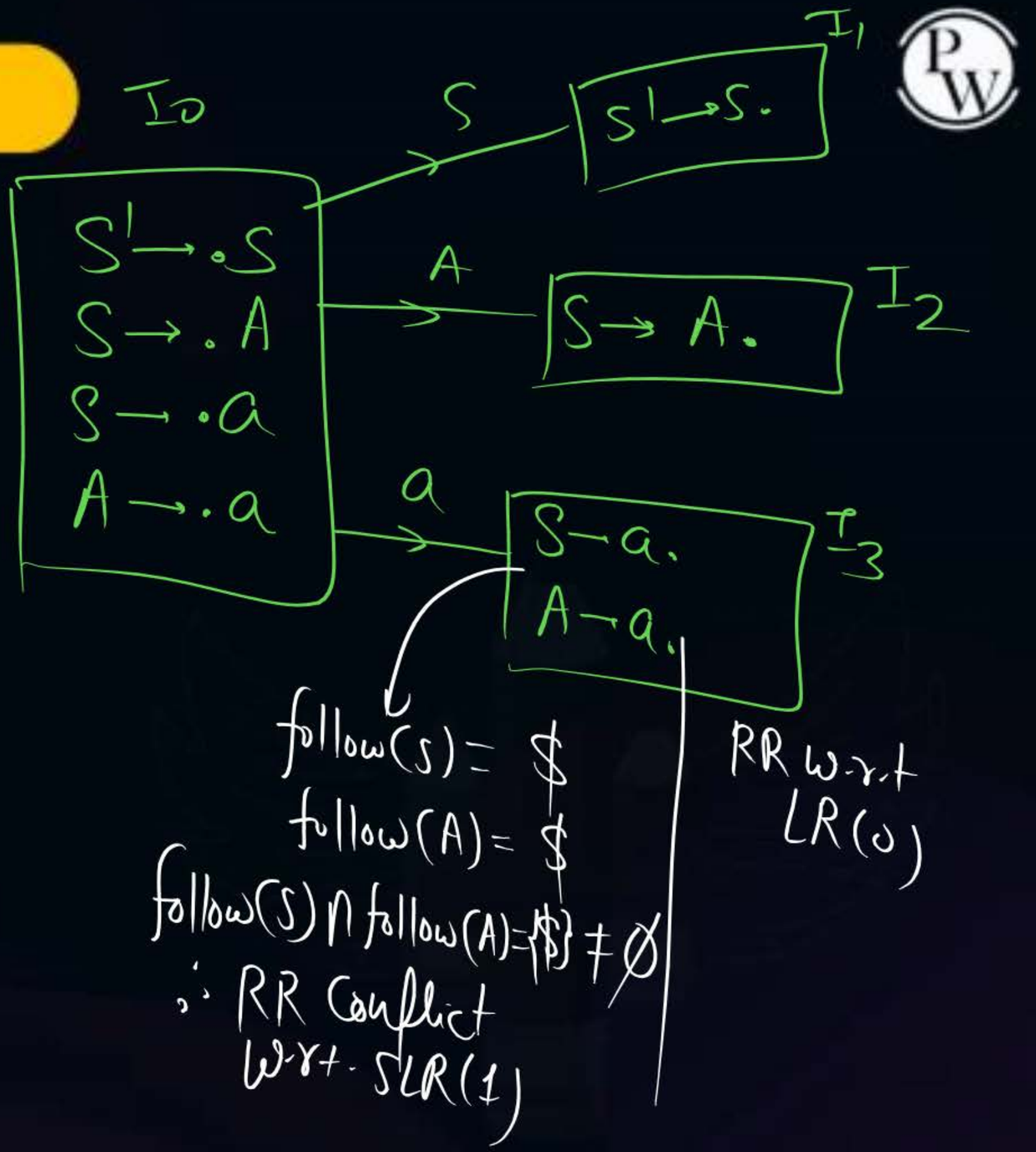


Topic : DFA Of LR(0) items

9. S→A / a

A→a

S'→S
S→A
S→a
A→a



Slide



Topic : LR(0) vs SLR(1)

- LR(0) Parser = LR(0) items + No Look ahead to place reduce entries in the parser table

- SLR(1) Parser = LR(0) items + Look ahead of length '1' is used to place reduce entries in the parser table



Topic : Is SLR(1) powerful enough

H.W.

Check whether the following grammar is SLR(1) or not?

S→Aa

S→dAb

S→dca

S→cb

A→c

Slide

Homework

$E \rightarrow E+T$

$E \rightarrow T$

$T \rightarrow T*F$

$T \rightarrow F$

$F \rightarrow (E)$

$F \rightarrow a$

Slide

Homework

$E \rightarrow E+T$

$E \rightarrow T$

$T \rightarrow TF$

$T \rightarrow F$

$F \rightarrow F^*$

$F \rightarrow a$

$F \rightarrow b$

Slide

Homework

$E \rightarrow E+T$

$E \rightarrow T$

$T \rightarrow TF$

$T \rightarrow F$

$F \rightarrow F^*$

$F \rightarrow (E)$

$F \rightarrow a$

$F \rightarrow b$

Slide



2 mins Summary

Topic LR(0) Parser

Topic SLR(1) Parser

Slide

THANK - YOU